Use of AI

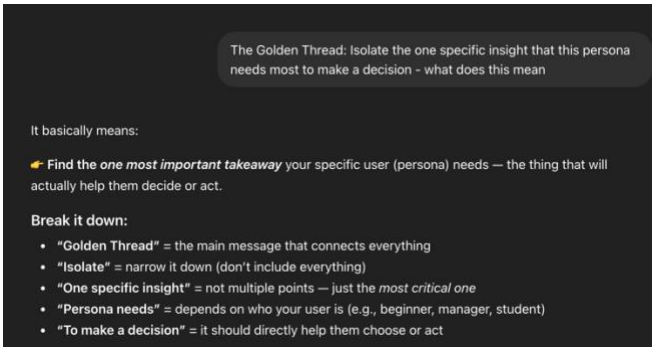
Declaration:

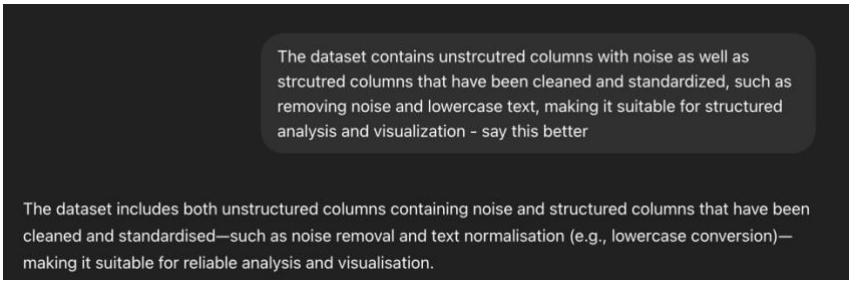
I acknowledge the use of ChatGPT to help me understand certain concepts and help refine my sentences. The output was modified by applying the learned concepts and I checked the accuracy of any information, data or sources generated by cross checking with online sources.

Justification:

I used ChatGPT to help me better understand some of the concepts used in this assignment such as “the Golden thread” and functionalities of Tableau as well. This helped me to apply these concepts correctly in my analysis. I also used ChatGPT to improve formality of words and help my writing, by refining sentences and vocabulary. This helped make my explanations clearer and more brief. Overall, I found ChatGPT useful as a support tool for learning unfamiliar concepts and improving the quality of my writing.

Below I have attached screenshots for example.





Link to transcript:

https://chatgpt.com/share/69d4f805-d694-83a1-aa7e-0e5c4392b403

(Next page is the transcript attached)

The Golden Thread: Isolate the one specific insight that this persona needs most to make a decision - what does this mean

It basically means:

👉 **Find the *one most important takeaway*** your specific user (persona) needs — the thing that will actually help them decide or act.

Break it down:

- **"Golden Thread"** = the main message that connects everything
- **"Isolate"** = narrow it down (don't include everything)
- **"One specific insight"** = not multiple points — just the *most critical one*
- **"Persona needs"** = depends on who your user is (e.g., beginner, manager, student)
- **"To make a decision"** = it should directly help them choose or act

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Simple example:

If your persona is a **busy teacher** looking at misconceptions:

❌ Too much info:

- "There are many misconceptions across categories, with varying frequencies..."

✅ Golden Thread:

- **"Focus on Topic X — it has the highest number of widely believed misconceptions, so teaching this first will have the biggest impact."**

In your assignment context:

You're basically answering:
👉 *"If they only remember ONE thing from this slide, what should it be?"*

Quick formula you can use:

“The most important thing for [persona] is that [key insight], because [impact on decision].”

Example:

> The most important insight is that a small number of misconceptions are highly prevalent, meaning prioritising these will maximise teaching impact.

If you want, tell me your persona and I can help you write a perfect “Golden Thread” line for your slide 👍

The dataset contains unstrcutred columns with noise as well as strcutred columns that have been cleaned and standardized, such as removing noise and lowercase text, making it suitable for structured analysis and visualization - say this better

The dataset includes both unstructured columns containing noise and structured columns that have been cleaned and standardised—such as noise removal and text normalisation (e.g., lowercase conversion)—making it suitable for reliable analysis and visualisation.

It uses a bubble like chart to represent the information of miscocneptions making it easier to understand and where, the numerical variable is encoded through bubble size to represent magnitude, while categorical variables are reflected through labels and grouping. This structure helps users to easily identify popular misconceptions in different groups at the same time. The bigger bubbles show the more popular ones overall and in each group its ordered from least popular to most popular. The dashboard only contains bubbles so it is easy to know these have to be interactive to get information, one its

hovered on a bubble the information relating to it aprreas like the category, misconcetion ,the accurate fact regarding it and the number of google hits the misconceton has. One example is given for the most poplaur misconcetiopn as well. The background o fht edahboard is blue and bubbles are white to make it good contrast to identify. - say this better

The dashboard uses a bubble chart to present misconceptions in a clear and intuitive way. Magnitude is represented by bubble size, while categories are conveyed through labels and grouping. This allows users to quickly identify the most common misconceptions both overall and within each category. The bubbles are arranged from least to most popular, making comparisons straightforward.

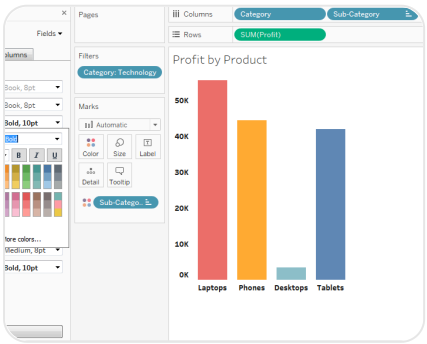
The design is simple and focused, using only bubbles to encourage interaction. When a user hovers over a bubble, additional details appear, including the category, the misconception, the correct fact, and the number of Google hits. A highlighted example is also provided for the most popular misconception.

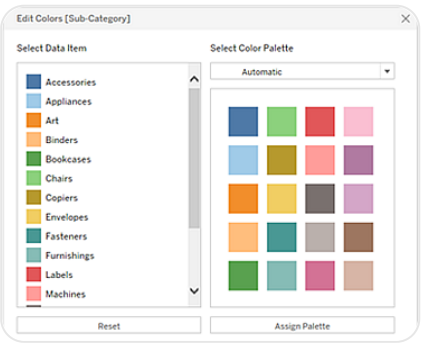
The blue background with white bubbles creates strong visual contrast, improving readability and making key information easy to identify.

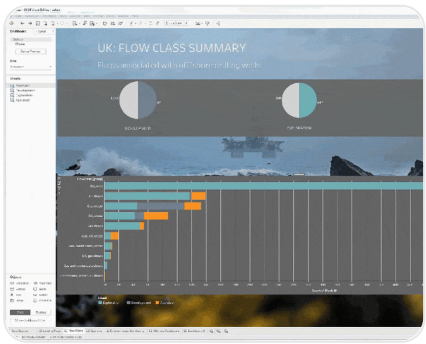
how to change background color on tabeau

Changing the background color in Tableau depends on *what exactly* you want to style (worksheet vs dashboard). Here's the quick guide:

🎨 1. Change Worksheet Background (your chart area)

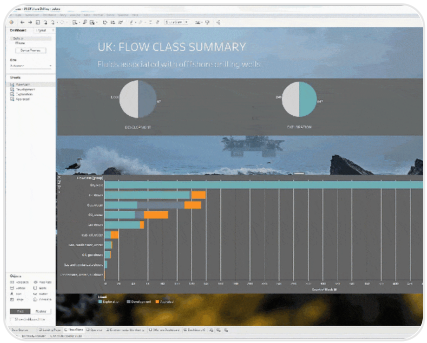


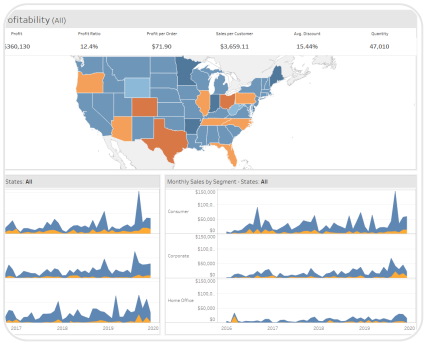
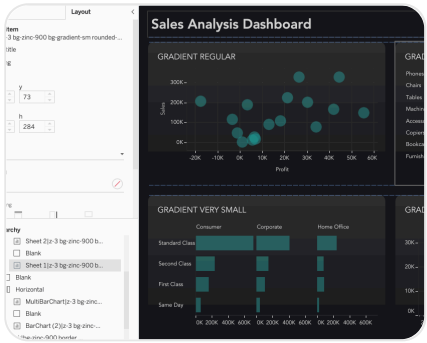




1. Open your worksheet
2. Go to **Format** (top menu)
3. Click **Shading**
4. Under:
   - **Worksheet** → changes the overall background
   - **Pane** → changes the chart area
5. Choose your desired color

🧩 2. Change Dashboard Background



1. Open your dashboard
2. Click anywhere on the blank dashboard area
3. In the **Layout pane (left side)**:
   - Find **Background**
   - Select your color

👉 You can also:

- Click a **container or object**
- Change its background separately for more design control

✨ Pro Tips

- Use **dark background + light marks** (like your blue + white bubbles idea) for strong contrast
- Keep colors consistent across worksheets and dashboard
- Avoid very bright backgrounds—they reduce readability

If you want, I can suggest the *best color combo* for your bubble chart to make it look more professional 👌

what are heat map annotations

Heat map annotations are **visual labels or highlights added on top of a heat map** to make key patterns easier to understand.

🔍 First, what's a heat map?

A heat map uses **color intensity** (e.g., red = high, blue = low) to show values or importance.

🧠 So what are annotations?

Annotations = extra information layered on top, such as:

- Text labels (e.g., "High confusion here")
- Numbers (actual values inside cells)
- Arrows or callouts
- Highlighted zones (e.g., red/yellow areas)

📊 Example (in your assignment context)

If you're using a heat map to show **cognitive load or confusion areas**:

- 🔴 Red zone → "Users struggle here"
- 🟡 Yellow zone → "Moderate confusion"
- 🟢 Green zone → "Easy to understand"

👉 The **colors = heat map**
👉 The **labels/arrows/explanations = annotations**

💡 Why use heat map annotations?

They help:

- Guide the viewer's attention
- Explain *why* something is important
- Make insights clearer (not just visual, but meaningful)

🧾 Simple one-line definition (you can use this):

Heat map annotations are labels or visual markers added to a heat map to highlight and explain key patterns, areas of intensity, or important insights.

If you want, I can help you design **red/yellow/green annotations for your slide** so it looks more professional 👍

A teacher may find it difficult to immediately identify which misconceptions are most important, as there is no order for popular misconceptions overall but just displayed by groups randomly. - say this better

A teacher may find it difficult to quickly identify the most important misconceptions, as they are not ranked by overall popularity and are instead presented within groups without a clear order.